

Maxwell Lo

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TECHNICAL SKILLS

Languages: Python, TypeScript, Java, SQL

Frontend: React, Next.js, Material UI

Backend & Infra: AWS (Lambda, RDS), PostgreSQL, Redis, OpenAI API

WORK EXPERIENCE

Junior Software Engineer | C3.ai (Contract via Paradyme) | Redwood City, CA **Jan 2025 - Aug 2025**

- Delivered Generative AI pilots for diverse clients by developing scalable backend solutions in Python, JavaScript, and C3.ai's proprietary type system, and enhancing UIs with React and UISDL.
- Designed and configured pipelines for data ingestion, multimodal file chunking, and email integrations.
- Served as lead software engineer on Montana Department of Revenue GenAI Pilot, driving architecture decisions, deployments, and cross-functional collaboration.

Software Development Intern | Beecoming | Berkeley, CA **May 2022 - Aug 2022**

- Developed microservices in JavaScript, exposing REST APIs to query information on prospective colleges.
- Conducted market research and designed new product features, including a feature to personalize college and course selections based on user preferences.
- Implemented automated Python scripts to minimize excess data stored in AWS S3 buckets.

Lead Computing Services Officer | Eta Kappa Nu (EECS Honor Society) | Berkeley, CA **Jan 2022 - Dec 2023**

- Led development and maintenance of internal web tools using Ruby and Python (Django), improving course survey data accessibility for students and faculty.
- Coordinated engineering work via Github issues and code reviews, enhancing workflow efficiency.
- Managed authentication, permissions and platform reliability for 100+ active users per semester.

PROJECTS

Farming Simulator | <https://react-virtual-garden.vercel.app> **2025**

- Built a browser-based farming simulator using React + TypeScript with a serverless Next.js + AWS Lambda + PostgreSQL (RDS) backend and Firebase-based authentication.
- Implemented server-authoritative game logic with time-based crop growth for offline progression.
- Designed a modular, object-oriented domain model with normalized schemas and reliable state serialization.

Baking Ingredient Substitutor | <https://ingredient-substituter.vercel.app/> **2025**

- Designed and implemented a full stack Next.js application using React, TypeScript, and Material UI to give ingredient substitution recommendations and their effects on baked goods.
- Built a deterministic substitution engine combined with GPT-generated explanations for baking analysis.
- Integrated OpenAI APIs with server-side validation, Redis caching, and IP-based rate limiting to reduce LLM usage and control costs.

EDUCATION

B.S. Electrical Engineering and Computer Science | University of California, Berkeley | GPA: 3.5/4.0 **Dec 2023**